

A 2-year follow-up of rehabilitation after ACL reconstruction using patellar tendon or hamstring tendon grafts: a prospective randomised outcome study

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Abstract Sixty-eight patients were clinically evaluated preoperatively, 3, 5, 7, 9 months, 1 and 2 years after ACL reconstruction, 34 with patellar tendon graft, 34 with hamstring graft. Outcome regarding graft choice and anterior knee laxity ($P = 0.04$) was in favour of patellar tendon graft. Hamstring graft led to a larger laxity, 2.4 mm compared with patellar tendon graft, 1.3 mm at 1 year and 2.5 mm and 1.5 mm, respectively, at 2 years ($P = 0.05$). There was a significant difference in rotational knee stability in favour of the patellar tendon graft at all test occasions but 9 months. A general effect regarding graft choice and muscle torque was found at 90°/s for quadriceps ($P = 0.03$) and hamstrings ($P \leq 0.0001$) and at 230°/s for hamstrings ($P \leq 0.0001$). No treatment effect regarding graft choice and one-leg hop test, postural sway or knee function was found. No group differences in anterior knee pain were found at any of the test occasions but 2 years in favour of hamstring graft compared to patellar tendon graft ($P = 0.04$). Patellar tendon graft resulted in higher activity level than hamstring graft at all test occasions but 1 year

($P = 0.01$). Patellar tendon ACL reconstruction led to more stable knees with less anterior knee laxity and less rotational instability than hamstring ACL reconstruction. Hamstring graft patients had not reached preoperative level in hamstring torque even 2 years after ACL reconstruction. Athletes with patellar tendon graft returned to sports earlier and at a higher level than those with hamstring graft.

Keywords ACL rehabilitation · Knee laxity · Muscle strength · Subjective outcome

Introduction

Anterior cruciate ligament (ACL) injury is one of the most serious injuries related to sports performance. This type of knee injury is especially common in athletes participating in so called pivoting sports that are characterised to put high demands on knee joint stability. No consensus exists about the treatment, although most orthopaedic surgeons suggest that the ruptured ACL in athletes should be treated with reconstructive surgery when considering return to pivoting sports. During recent years it has been a shift from the use of patellar tendon to hamstring tendons in terms of graft choice for ACL reconstruction. However, the graft used for an optimal clinical outcome still remains controversial [12, 31]. When it comes to rehabilitation most authors have described the use of a similar rehabilitation programme irrespective of graft [2, 6, 8, 10, 18]. It should be pointed out though, that evidence is still lacking concerning how the different grafts influence physical outcome over time. Therefore, the aim of the present investigation was to evaluate both a short and a 2-year follow-up of physical outcome in patients who have received the same rehabilitation after ACL reconstruction with either patellar

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tendon or hamstring tendon grafts. It was hypothesised that, irrespective of graft choice, the patients were fully recovered in terms of hamstring muscle torque at the 2-year follow-up.

Materials and methods

Patients

Between 1999 and 2005, 80 ACL injured patients, 42 males and 38 females met the inclusion criteria and were involved in preoperative physiotherapy at a Sport Rehabilitation Clinic. Six males and six females declined to participate due to lack of time ($n = 8$), no ACL reconstruction was performed ($n = 3$), and transferral to another city ($n = 1$). Sixty-eight patients, 36 males and 32 females were finally included as participants in the study (Table 1).

The inclusion criteria were age between 16 and 50 years and a symptom free contra-lateral knee. Patients with a medial or lateral meniscus tear and/or a medial collateral ligament injury grade I, where surgical repair was not indicated, were also included. The exclusion criteria were injuries other than the above mentioned, uncorrected vision, a medial collateral ligament injury grade II or III, meniscus lesions suitable for fixation, and patients operated with patellar tendon or hamstring grafts harvested from their contra-lateral leg.

Table 1 Demographic data for patients with patellar tendon bone-to-bone (PTBTB) graft ($N = 34$) and patients with hamstring graft ($N = 34$)

	PTBTB graft ($n = 34$)	Hamstring graft ($n = 34$)
Males/females	22/12	14/20
Age, years (M, SD)	29 (7)	30 (9)
Height, m (M, SD)	1.75 (0.08)	1.73 (0.09)
Weight, kg (M, SD)	75.1 (10.1)	72.4 (11.9)
BMI, kg/m ² (M, SD)	24.3 (2.4)	24.2 (2.9)
Injured leg left/right	15/19	13/21
Median time between injury and surgery, months (lower and upper quartile)	7.8 (5–12)	8.5 (5–18)
Semitendinosus graft alone (n)		23
Semitendinosus + gracilis graft (n)		11
Medial meniscus injury	11	7
Lateral meniscus injury	6	8
Patella cartilage damage	3	4
Tibia cartilage damage	3	3
Femur cartilage damage	7	9
Medial collateral ligament injury	2	1

Twenty experienced orthopaedic surgeons performed the ACL reconstructions with patellar tendon or hamstring grafts. Four patients, who had undergone reconstruction with hamstring graft, and four patients, who had undergone reconstruction with patellar tendon graft, had to undergo another arthroscopy during the follow-up period. The reasons for this were; removal of a shaver blade that was lost during the index operation ($n = 1$), extraction of a “loose body” in the knee joint ($n = 1$), pain in the knee joint ($n = 3$), trauma, suspected graft rupture ($n = 1$), removal of cyclops syndrome ($n = 1$), resection of extensive scar tissues in the knee joint ($n = 1$).

Group randomization and rehabilitation

This study is a follow-up study based on an earlier publication [17] aimed to investigate whether early introduction of closed kinetic chain exercises for quadriceps influenced early outcome after ACL reconstruction with either patellar tendon graft or hamstring graft. In that study, stratified for gender, the patients were postoperatively randomised into four different groups in terms of rehabilitation. At the first postoperative rehabilitation session each patient with patellar tendon graft or hamstring graft, respectively, was asked to choose one out of 50 closed envelopes (25 for males and 25 for females) for group randomization. At the end of the study there were three envelopes left for males and 13 for females with patellar tendon graft and 11 envelopes left for males and five for females with hamstring graft. The reason for using more envelopes than was calculated by the power analysis was that patients included late in the study should have the same chance to be randomised to either of the groups. The earlier reported difference in terms of rehabilitation for each type of graft could not be seen when analysing the data reported in the present study. Therefore, only the outcome based on type of graft is reported here.

All patients started the standardised postoperative rehabilitation programme within 1 week after surgery at the same outpatient clinic. Supervised physiotherapy was performed 2–3 times a week as long as the patient and the physiotherapist considered it necessary. The median number of training sessions was 51 (18–109) for the patients reconstructed with patellar tendon graft and 50 (13–93) for the patients reconstructed with hamstring graft. The rehabilitation protocol consisted of joint and muscle flexibility exercises, balance- and coordination training and strength training focusing mainly on the thigh muscles. One group of patients with patellar tendon ACL reconstruction started with open kinetic chain quadriceps exercises 4 weeks postoperatively, while another group started 12 weeks postoperatively. The same goes for patients operated on with hamstring tendon grafts. No brace was used during the

rehabilitation period. Immediate weight bearing according to tolerance was allowed after surgery. Based on muscle strength, balance/coordination and functional performance the patients were allowed to return to active, competitive sports 6 months postoperatively or later depending on the functional capacity. Full description of the rehabilitation protocol can be seen in an earlier publication [17]. Ten experienced physiotherapists were involved in the rehabilitation.

Postoperative ACL ruptures

One female patient sustained a re-rupture (hamstring tendon graft) of her reconstructed knee and one male patient sustained a new ACL injury of his contra-lateral knee within 2 years after the first ACL reconstruction. These injuries are included in the statistical calculations using the mixed model statistical method.

Evaluation

All patients were evaluated by two experienced independent examiners within 4 weeks prior to surgery as well as 3, 5, 7, 9 months, 1 and 2 (mean 24.7 months, SD 1.2) years after surgery. None of these two examiners were involved in the rehabilitation. The same examiner followed the same patient at all his/her test occasions and both examiners were blinded for type of rehabilitation (early or late start of open kinetic chain quadriceps training), but not for type of reconstruction.

The evaluation consisted of the methods presented below. All measurements at each test occasion were carried out in the following order.

Anterior knee laxity

Anterior tibial displacement relative to the femur was measured bilaterally at approximately 20° of knee flexion with the KT-1000 arthrometer (MEDmetric, Corp., San Diego, CA, USA) at max manual. The difference in displacement between the operated and the contra-lateral uninjured knee was expressed in millimetres [7, 30].

Pivot shift

The pivot shift test was used to control possible rotational instability and recorded according to the International Knee Documentation Committee (IKDC) [15, 16].

Thigh muscle torques

The Kin-Com[®] dynamometer (Chattecx Corp., Chattanooga, TN, USA) [9] was used for measuring concentric and

eccentric muscle torques of the quadriceps and hamstring muscle groups at 90°/s as well as 230°/s and within 90°–10° of knee flexion, always starting with the contra-lateral uninjured leg. At the 3-month test occasion, thigh muscle torque was only tested within 90°–40° of knee flexion.

One-leg hop test for distance

The one-leg hop test for distance was used for functional hop performance [5]. The patient was standing on one leg and instructed to jump straight ahead as far as possible. Three trials were performed on both legs, and the patient decided which leg to start with [5]. The best trial of each leg was chosen for statistical calculations. The ratio between the legs was calculated and reported in percent.

Postural sway

Kinesthetic Ability Trainer 2000 (KAT 2000TM) (BREG Inc., CA, USA) was used for measuring balance/postural sway when standing on one leg. KAT 2000 has been found to be reliable for testing groups of individuals [14]. The measurements were performed during standardised conditions by having the patient standing bare-foot on one leg, with that knee as well as the knee of the free leg flexed approximately 30°, and the arms crossed in front of the chest. Three tests, 20 s each, were performed on both legs starting with the uninjured contra-lateral leg. The distance between the centres of the board to centre of pressure was measured in arbitral units. The best trial (least postural sway) was used for statistical calculations.

Anterior knee pain

An anterior knee pain (AKP) score modified from an earlier publication [29] was used to evaluate possible AKP. This score has been revised, adjusted and specified for possible AKP in ACL injured patients (Werner S, unpublished data). The score consists of 8 subgroups; pain, occurrence of pain, walking upstairs, walking downstairs, sitting with flexed knee >30 min, squatting, kneeling and arretations—catching. 50 points is the maximum score which is equal to no AKP. The sensitivity of the score was studied on patients with ACL injuries, patients after patellar tendon ACL reconstruction, patients after hamstring tendon ACL reconstructions, patients with postoperative meniscus lesions, and uninjured controls. With regards to the total score there was a significant difference between uninjured controls and ACL injured patients ($P = 0.0001$). A test–retest procedure indicated good reliability with an intra-class correlation (ICC) of 0.97 of the total score (Werner S, unpublished data).

Knee injury osteoarthritis outcome score (KOOS)

The Knee injury osteoarthritis outcome score (KOOS) was used to assess the patient's opinion about his/her knee function and possible associated problems. [23, 24]. KOOS consists of 42 items, grouped into five separate subscales; (1) pain, (2) other symptoms, (3) function in daily living (ADL), (4) function in sport and recreation (Sport/Rec) and (5) knee-related quality of life (QoL).

Tegner activity scale

The Tegner activity scale [28], which describes the patient's type as well as level of activity (0–10) was used. A high score means that the patient participates in sports that place high demands on knee joint stability.

The present investigation has received approval from the Ethics Committee at the Karolinska Institutet, Dnr 99-091. All patients granted their informed consent prior to participation.

Statistical methods

Prior to the study a power analysis was made based on anterior knee laxity. A difference of 1.5 mm side-to-side difference in anterior knee laxity of the reconstructed knee estimated a sample size of 15 patients in each group with 80% power when $P \leq 0.05$. The power analysis was made as a "two-group repeated measures ANOVA" (Greenhouse-Geisser correction) in the nQuery 4.0. Due to the multiple methods used in this study, more patients were included.

The present study has a longitudinal design with two parallel treatment groups. The demographic data of the patients and the training volume are presented in means and standard deviations or medians and lower and upper quartiles.

In terms of testing the AKP score for reliability and sensitivity, repeatability was measured by an ICC. Group differences were analysed by one-way ANOVA and post hoc comparisons were made according to the Tukey honestly difference test. If the distributions were severely skewed, repeatability was measured by Mann–Whitney *U*-test and Kendall's rank order correlation and group differences by Kruskal–Wallis test and Mann–Whitney *U*-test. The data from the AKP score, pivot shift and KOOS were considered as non-parametric data and Mann–Whitney *U*-test was used when calculating the results.

In order to compare the ratio in thigh muscle torques, differences in anterior knee laxity and differences in postural sway between the operated knee and the uninjured contra-lateral knee over time, a mixed effect model analysis was used. These analyses included testing for graft

choice, the effect of concentric and eccentric muscle actions, a time effect and the effect of whether the ACL reconstruction was performed in the left or right knee. Further, the analyses for anterior knee laxity included a covariance structure called compound symmetry.

In situations in which a significant general treatment effect was observed, planned pair-wise comparisons were used. To adjust the significant levels of the planned comparisons, a Bonferroni correction was made. SAS PROC MIXED was used for these analyses.

Results

Anterior knee laxity

A general treatment effect was found in terms of graft choice and laxity ($P = 0.04$). For the 3, 5, 7 and 9 months follow-up no general treatment effect was found. At the 1-year follow-up a statistical significant difference ($P = 0.03$), in laxity was found between the operated and the uninjured knee for the patellar tendon graft, 1.3 mm (SD 0.4) and for the hamstring graft, 2.4 mm (SD 0.3). At the 2-year follow-up it was 1.5 mm (SD 0.3) and 2.5 mm (SD 0.4), respectively ($P = 0.05$) (Fig. 1).

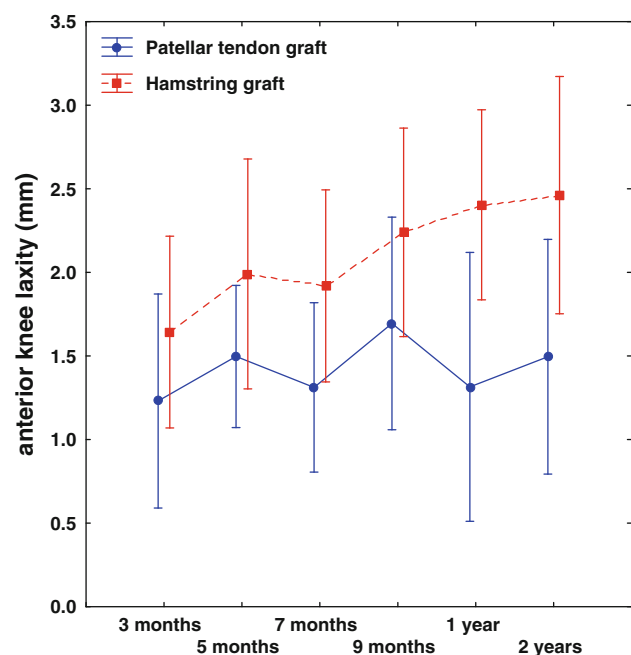


Fig. 1 Anterior knee laxity (mm) (mean and CI) between the healthy- and the ACL-reconstructed knee at 3, 5, 7, 9 months, 1 and 2 year follow-up for patients after patellar tendon ACL-reconstruction ($n = 34$) and hamstring ACL-reconstruction ($n = 34$)

Pivot shift

A general treatment effect was found in terms of graft choice and rotational knee stability ($P < 0.05$). A statistical significant difference between the patellar tendon graft and the hamstring graft, in favour of the patellar tendon ACL reconstruction, was seen at all follow-ups except for the 9-months follow-up (n.s.) (Table 2).

Muscle torques

A general treatment effect in terms of graft choice and muscle torque over time was found for quadriceps torque at 90°/s ($P = 0.03$), hamstring torque at 90°/s ($P < 0.001$) and hamstring torque at 230°/s ($P < 0.001$), but not for quadriceps torque at 230°/s (n.s.) (Figs. 2, 3, 4, 5).

One-leg hop test for distance

No general treatment effect in terms of graft choice and one-leg hop test for distance was found (n.s.) (Table 3).

Postural sway

The preoperative values were used as a covariate in the present analysis. There were no significant group differences in terms of postural sway over time (n.s.).

Anterior knee pain

No significant group differences in terms of AKP were found preoperatively or at any of the follow-ups except for the 2-year follow-up. A significant difference, in favour of the hamstring tendon graft, $N = 30$, 45.5 (28–50), compared to patellar tendon graft, $N = 30$, 42 (26–50) was observed at the 2-year follow-up ($P = 0.04$).

Knee injury osteoarthritis outcome score (KOOS)

No significant group differences regarding knee function and quality of life as determined with KOOS were found over time in neither of the subscales and therefore all patients were analysed together as one unit. A significant improvement irrespective of graft used was seen over time in all subscales ($P < 0.001$) (Fig. 6).

Tegner activity scale

No significant group differences in terms of activity level were found preoperatively or at 5, 7, 9 months or 2 years after ACL reconstruction. A significant difference, in favour of the patellar tendon graft (7, range 1–10 versus 5, range 2–8) was seen at the 1-year follow-up ($P = 0.01$).

Table 2 Numbers and percentage (%) of assessed patients with Pivot Shift preoperatively and at 3, 5, 7 and 9 months as well as 1 and 2 years postoperatively according to the IKDC classification (1 = none, 2 = +(glide), 3 = ++(clunk) and 4 = +++(gross) after patellar tendon or hamstring ACL-reconstruction

	Patellar tendon graft <i>n</i> (%)	Hamstring graft <i>n</i> (%)	<i>P</i> value
Assessed patients (<i>n</i>):	30	32	
Pre-op			
1	2 (6.5)	–	0.61
2	12 (40)	12 (38)	
3	14 (47)	18 (56)	
4	2 (6.5)	2 (6)	
Assessed patients (<i>n</i>):	32	30	
3 Months			
1	20 (62.5)	10 (33)	0.05
2	11 (34.5)	18 (60)	
3	1 (3)	2 (7)	
4	–	–	
Assessed patients (<i>n</i>):	26	29	
5 Months			
1	16 (61.5)	8 (27.5)	0.04
2	9 (34.5)	20 (69)	
3	1 (4)	1 (3.5)	
4	–	–	
Assessed patients (<i>n</i>):	27	27	
7 Months			
1	21 (78)	11 (40.5)	0.01
2	6 (22)	14 (52)	
3	–	2 (7.5)	
4	–	–	
Assessed patients (<i>n</i>):	22	26	
9 Months			
1	11 (50)	8 (31)	0.37
2	9 (41)	16 (61.5)	
3	2 (9)	2 (7.5)	
4	–	–	
Assessed patients (<i>n</i>):	31	30	
1 Year			
1	19 (61)	9 (30)	0.008
2	12 (39)	15 (50)	
3	–	6 (20)	
4	–	–	
Assessed patients (<i>n</i>):	30	29	
2 Years			
1	17 (57)	8 (27.5)	0.02
2	10 (33)	13 (44.5)	
3	3 (10)	7 (24)	
4	–	1	

The *P* values for the whole model is <0.05 . The *P* values for each follow-up are shown in the last column

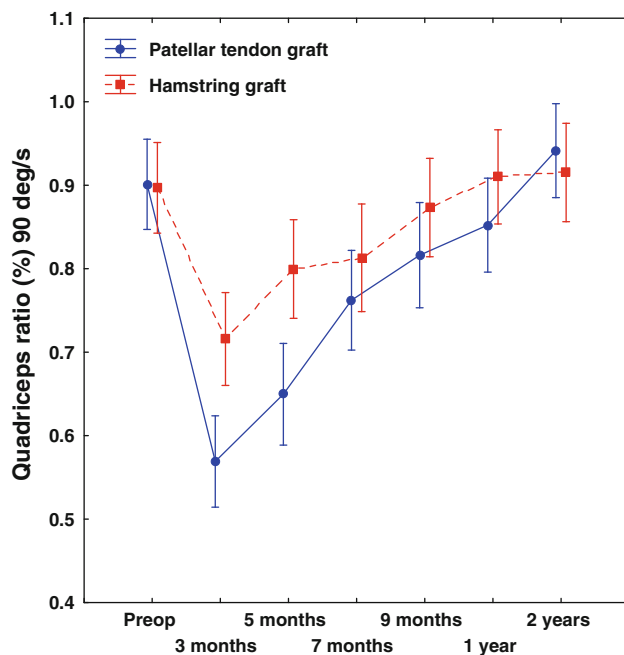


Fig. 2 The quadriceps muscle ratio at the velocity 90°/s, (ACL-reconstructed knee/healthy knee) (mean and CI) preoperatively, at 3, 5, 7, 9 months and 1 and 2 year follow-up for patients after patellar tendon ACL-reconstruction ($n = 34$) and hamstring ACL-reconstruction ($n = 34$). *At the 3-month follow-up the test was performed within 90°–40° of knee flexion

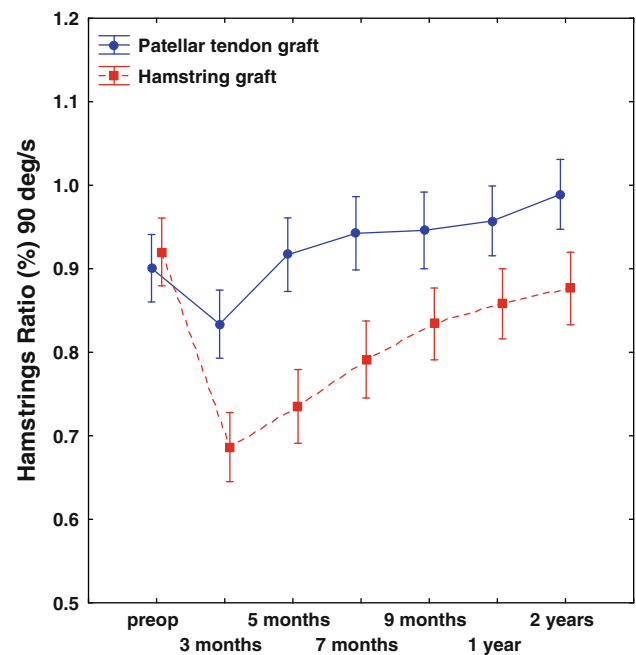


Fig. 4 The hamstring muscle ratio at the velocity 90°/s, (ACL-reconstructed knee/healthy knee) (mean and CI) preoperatively, at 3, 5, 7, 9 months and 1- and 2-year follow-up for patients after patellar tendon ACL-reconstruction ($n = 34$) and hamstring ACL-reconstruction ($n = 34$)

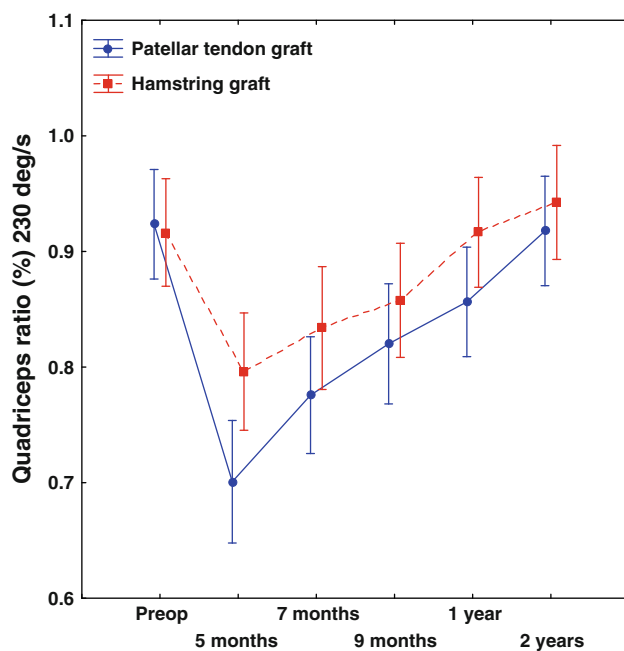


Fig. 3 The quadriceps muscle ratio at the velocity 230°/s, (ACL-reconstructed knee/healthy knee) (mean and CI) preoperatively, at 5, 7, 9 months and 1- and 2-year follow-up for patients after patellar tendon ACL-reconstruction ($n = 34$) and hamstring ACL-reconstruction ($n = 34$)

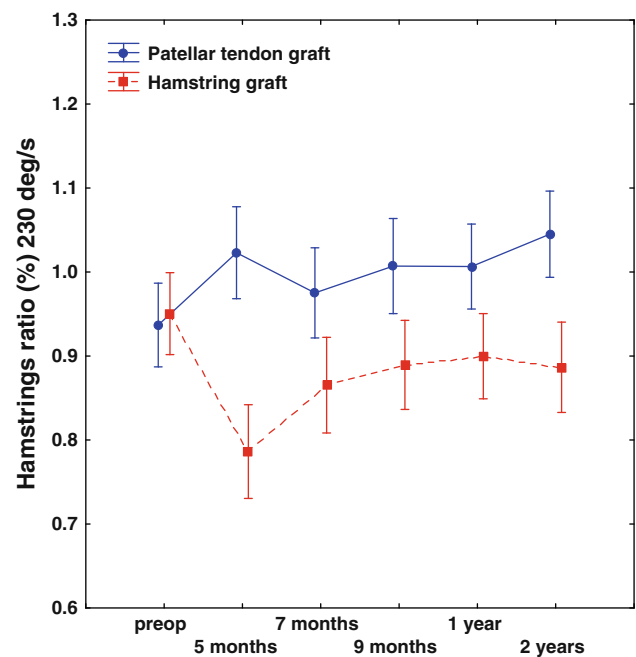


Fig. 5 The hamstrings muscle ratio at the velocity 230°/s, (ACL-reconstructed knee/healthy knee) (mean and CI) preoperatively, at 5, 7, 9 months and 1- and 2-year follow-up for patients after with patellar tendon ACL-reconstruction ($n = 34$) and hamstring ACL-reconstruction ($n = 34$)

Table 3 Descriptive data from the one-leg hop test (mean and SD) and number of patients tested at each follow-up for patients reconstructed with patellar tendon graft and patients reconstructed with hamstring grafts. The ratio between the reconstructed knee/healthy knee is presented

	PTBTB graft	Hamstring graft
7-Month follow-up	0.85 (0.14) <i>N</i> = 22	0.91 (0.10) <i>N</i> = 16
9-Month follow-up	0.91 (0.11) <i>N</i> = 23	0.88 (0.14) <i>N</i> = 25
1-Year follow-up	0.91 (0.09) <i>N</i> = 29	0.94 (0.08) <i>N</i> = 29
2-Year follow-up	0.94 (0.07) <i>N</i> = 28	0.99 (0.07) <i>N</i> = 25

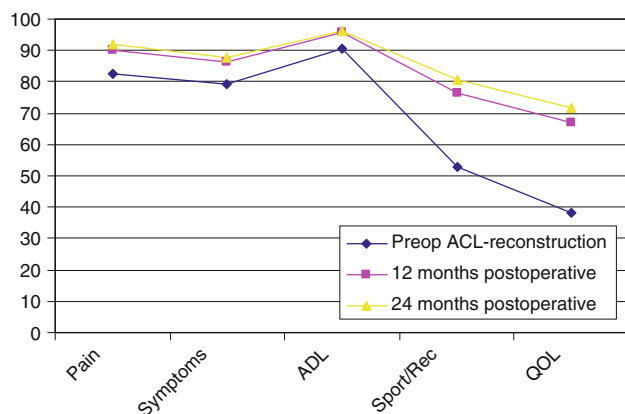


Fig. 6 Knee function according to knee injury osteoarthritis outcome score (KOOS) for the patients preoperatively (*N* = 68), 12 months (*N* = 64) and 24 months (*N* = 61) after ACL-reconstruction. *QOL* quality of life, *ADL* activity of daily living

Discussion

The principal and most interesting finding of the present study was that ACL reconstruction with hamstring graft led to somewhat greater laxity by time when compared with patellar tendon ACL reconstruction. Another finding was that 2 years after ACL reconstruction patients operated on with hamstring graft showed both significantly lower quadriceps and hamstrings strength ratio in comparison with patients operated on with patellar tendon graft as well as when compared with their own preoperative values.

From a surgical point of view a successful ACL reconstruction depends on several factors, for example the knee flexion angle and the graft of tensioning at the time of fixation. In a recent published review [3] it was reported that no randomised controlled studies have been performed in order to determine the most effective amount of tension to apply to hamstring grafts. The patients in the present study were operated on with “the same technique” by several orthopaedic surgeons but with similar experience of ACL reconstructions. However, the initial graft tension applied during fixation is not known. Whether the number of different orthopaedic surgeons has any influence on the result or whether it plays any significant role for functional

knee joint stability could be questionable, though. A further limitation of this study is that the choice of graft was not randomised for, but based on the surgeons’ preferences, for instance experience and patient-related factors such as gender or level of activity.

Moreover, this study was stratified for gender resulting in similar amount of males and females in each group, which could have influenced the results since it has been reported [11] that females, reconstructed with hamstring tendon graft, shows a significantly greater knee laxity compared to males. Laxdal et al. [18] reported no significant differences in terms of knee laxity in males operated on with patellar tendon graft or hamstring graft. Unfortunately, the cohort in the present study was too small to allow gender-related comparisons. The difference in anterior knee laxity and rotational stability between grafts found in the present study might be judged as small and questionable from a clinical point of view. But it could be speculated that in some sports with heavy demands on rotational and anterior stability and even a small side-to-side difference in anterior knee laxity might play an important role. Furthermore, it has theoretically been proposed that excessive tibial rotation will lead to osteoarthritis [26]. In a recent publication, Lidén et al. [19] reported degenerative changes (Fairbank system) of the knee joint in 74% of the patients who had undergone an ACL reconstruction with either patellar tendon graft or hamstring tendon grafts. It has been reported previously that patellar tendon is a more stiff tissue than hamstring tendon [21]. It is probable that this is one explanation for the somewhat higher side-to-side difference in anterior knee laxity and greater knee instability according to the pivot shift test after hamstring ACL reconstruction when compared with patellar tendon ACL reconstruction.

Thigh muscle strength is often reported to be one of the cornerstones of importance after ACL reconstruction [22]. The quadriceps muscle dynamically stabilises the knee joint in most functional activities but hamstrings play an important role in terms of shear forces by preventing the tibia from sliding anteriorly relative to the femur. After hamstring ACL reconstruction it took less than 1 year to reach preoperative level of quadriceps torques and not even 2 years after reconstruction they had reached preoperative levels of hamstring torques as evaluated with isokinetic dynamometry at both angular velocities used. After patellar tendon ACL reconstruction it took only 5 months to reach preoperative level of hamstring torques but 2 years to reach preoperative level of quadriceps torques irrespective of angular velocities used. Whether preoperative values is sufficient enough when planning to return to sport can be discussed since the patients usually have been injured for a time and therefore probably have less than normal strength. In terms of muscle qualities there are reasons to believe

that muscle power is the most important strength variable in sports [27]. Therefore, it may be considered a limitation in the present study that this type of muscle quality evaluation was not carried out. However, the results from our study is in agreement with a recently published study [1], reporting results from muscle power strength test in patient operated on with either patellar tendon or hamstring tendon grafts. Ageberg et al. [1] reported that patients operated on with hamstring tendon grafts had a significantly lower limb symmetry index in knee flexion power compared to those patients operated on with patellar tendon graft. This means that the hamstring muscles are considerably more vulnerable than the quadriceps muscles after hamstring ACL reconstruction. Therefore, these findings suggest that extra effort should be placed on strengthening the hamstrings after ACL reconstruction with hamstring graft. It also means that a safe return to sport earlier than 2 years after ACL reconstruction may be questionable since the hamstring muscles are regarded to be the major dynamic stabilisers in terms of shear forces in the knee joint [25].

There were no group differences with respect to the one-leg hop test for distance and postural sway. This indicates that these tests may not be sensitive enough to detect differences in performance due to graft choice. Earlier studies have pointed out the importance of using a battery of functional tests in order to improve sensitivity [13, 20]. Furthermore, a standardised test design by measuring the patient during both non-fatigued and fatigued conditions have been proposed in order to receive a reliable result of clinical relevance [4].

Feller and Webster [10] reported significantly more AKP after patellar tendon ACL reconstruction than hamstring tendon ACL reconstruction as measured with Visual Analogue Scale (VAS). Clinical suggestions have arisen that this might be an important factor in deciding which graft is the most appropriate one for a particular patient. We did not find any group differences in terms of AKP preoperatively and at all postoperative test occasions except at the 2 years follow-up, when patients with hamstring tendon ACL reconstructions reported somewhat less AKP than patients with patellar tendon ACL reconstructions. However, this small group difference most likely does not have any clinical relevance meaning that the risk of developing AKP after ACL reconstruction is probably not a reason for the choice of graft.

No significant group differences were seen over time regarding knee function and quality of life according to KOOS. These findings show that patellar tendon and hamstring tendon grafts are equally good in terms of subjective knee evaluation of knee function and quality of life. It should be pointed out, though, that irrespective of graft choice for ACL reconstruction the patients did not reported themselves as fully symptom free as judged by all the

different subgroups of KOOS. This is observed especially when it comes to sports and recreation as well as quality of life. However, this may be the natural course after ACL reconstruction.

Participation in sports according to Tegner showed a significant group difference of two levels in favour of the patellar tendon group at the 1-year test occasion. However, no differences between groups were found at the other test occasions. One can therefore speculate that patients with patellar tendon ACL reconstructions are able to return to sports earlier and at a higher level than those operated on with hamstring tendon grafts. This result is likely to be of special interest when dealing with athletes. The somewhat smaller anterior knee laxity and/or the better knee joint stability as determined with the pivot shift test in patients with patellar tendon ACL reconstruction compared to patients with hamstring tendon graft may play a role for this group in returning to sport. A stable knee joint probably leads to less fear of and better sports performance than a more unstable knee joint.

During the last decade, there has been a shift from patellar tendon graft to hamstring tendon grafts for ACL reconstruction. A non-evidence-based unreflective imitation of the rehabilitation protocol used after ACL reconstruction with patellar tendon graft can be seen in clinical practise. In light of the present study it is suggested that extra effort should be placed on strengthening of the hamstring muscles after hamstring ACL reconstruction. This might lead not only to improved muscle strength and muscle balance but to some extent prevent graft elongation.

Conclusion

Patellar tendon graft leads to more stable knees with less anterior laxity and less pivot shift. Patients with hamstring ACL reconstruction need more hamstring strengthening exercises. Some of our data indicate that patients operated on with hamstring grafts might need another and slower rehabilitation protocol focusing more on hamstring strength than those with patellar tendon graft. Athletic patients with patellar tendon ACL reconstructions returned to sports earlier and at a higher level than those operated on with hamstring tendon grafts.

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